

2EI4 DP #3

DP #3: Transistor Amplifier

Jasmine Dosanjh – dosanj5 – 400531879

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Part 1: Theory

Transistor Choice, Amplifier Topology, and Calculations



Transistor Choice

Using the chosen amplifier topology, the transistor was chosen:

- Input signal Equation for CD MOSFET

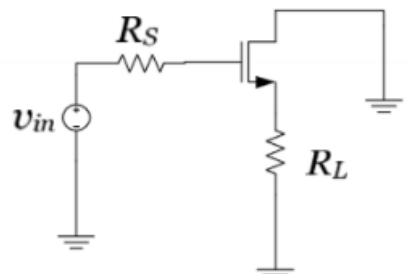
$$|V_{in}| \leq 0.2(V_{GS} - V_T)(1 + g_m R_L)$$

$$0.5 \leq 0.2(V_{GS} - V_T)(1 + 0.1g_m)$$

- g_m could not be found due to unknown V_{GS} .
- To find V_{GS} the current or R_S was needed.

$$V_{GS} = 0.5 - IR_S \rightarrow I = ? R_S = ?$$

- Therefore a BJT was used.

Common Drain

$\frac{g_m R_L}{1 + g_m R_L} \approx 1$
∞
$1/g_m$
$0.2(V_{GS} - V_T)(1 + g_m R_L) \geq V_{in}$

Topology Choice

Amplifier type	Gain	R _{in}	R _{out}	Signal swing	Application
CS/CE	High; inverting	M/H	H	Small or v. small	Main gain stage in most amplifier applications
CG/CB	High; non-inverting	V. low	H	Small or v. small	Intermediate gain stage; current buffer
CD/CC	< 1; can be made v. close to 1	V. high	V. low	M/H	Output stage; voltage buffer



Topology Choice

1) Signal swing

- Options 1 and 2 require $|V_{in}| \leq 5mV$ for high linearity.
- The design requires $|V_{in}| \leq 500mV$,
- Therefore, option 3 the only suitable topology

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Topology Choice

2) Voltage Buffer

- Design requires voltage loss $\leq 10\%$.
- CC amplifier is a voltage buffer; voltage gain is close to 1 . $\therefore 0.9V \leq \text{Gain} \leq 1V$
 - Therefore, $V_{out} \approx V_{in}$
 - Ensures signal is passed to the load without much loss.

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Calculations

Step 1) Find g_m

- Plug in known values into input signal Equation:

$$|V_{in}| \leq 0.2V_T(1 + g_m R_L)$$

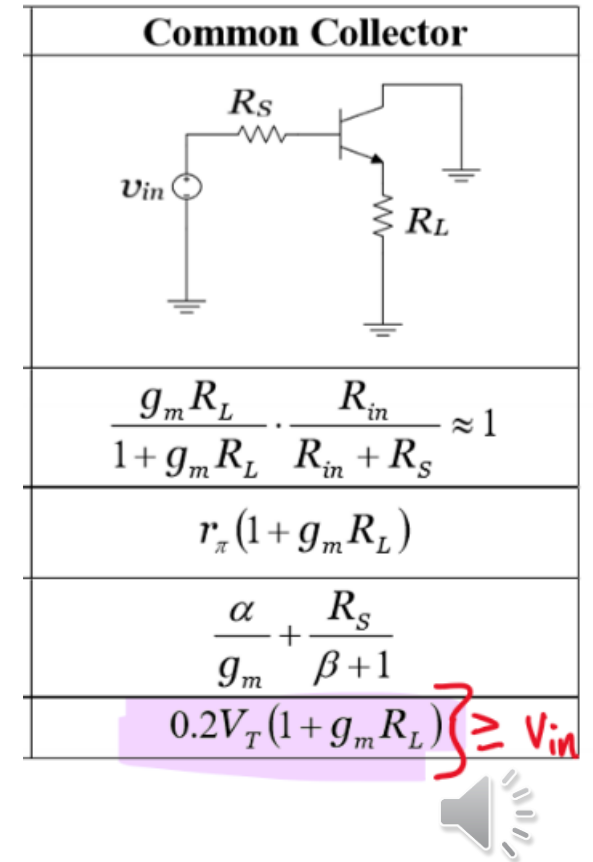
$$0.5 \leq 0.2(25mV)(1 + 0.1g_m)$$

$$\therefore 990mS \leq g_m$$

Step 2) Find bias-current

$$g_m = 40I_C$$

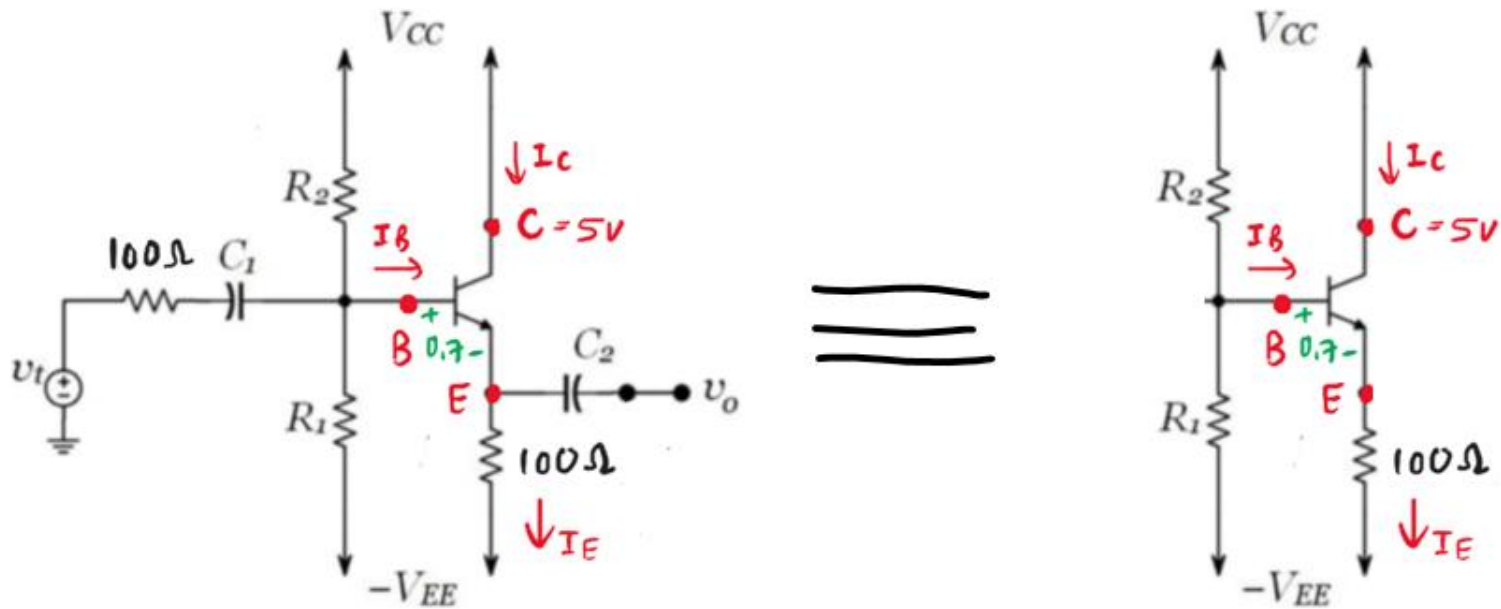
$$\therefore I_C = 24.75mA$$



Calculations

Step 3) Find Thevenin equivalent

- Let $V_{CC} = +5V$ and $V_{EE} = -5V$.
- In DC both capacitors are in steady state, so they are both open circuited:



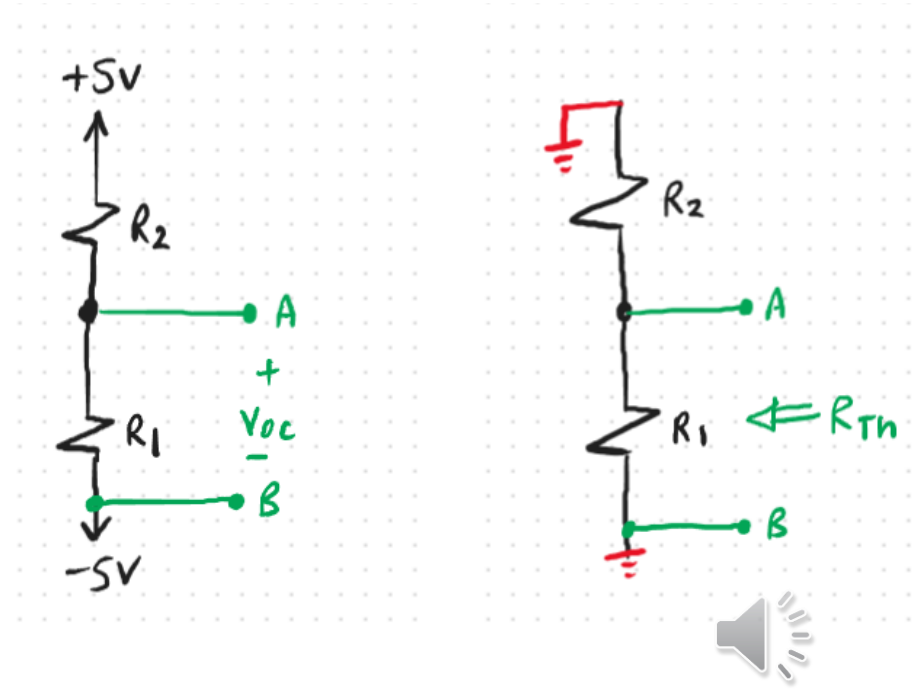
Calculations

Step 3) Find Thevenin equivalent

- Assume F.A. mode $\rightarrow V_{BE} = 0.7, I_C = \beta I_B$
- Thevenin Equivalent to left of V_B :

$$V_{Th} = -5 + 10 \left(\frac{R_2}{R_1 + R_2} \right)$$

$$R_{Th} = R_1 \parallel R_2$$



Calculations

Step 3) Find Thevenin equivalent

- Apply KVL (at green path):

$$-5 + 10 \left(\frac{R_2}{R_1 + R_2} \right) - I_B (R_1 \parallel R_2) - 0.7 - 0.1 I_E = -5$$

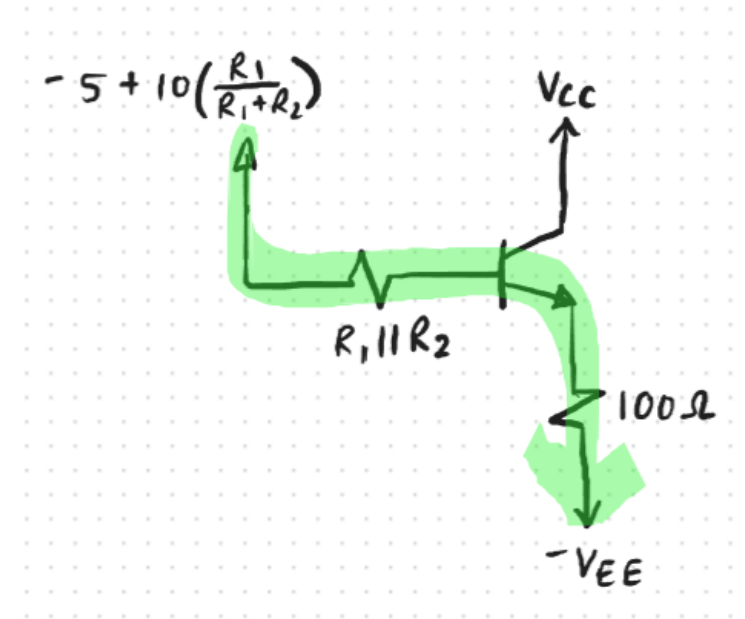
- Let $V_{BB} = -5 + 10 \left(\frac{R_2}{R_1 + R_2} \right) \rightarrow (1)$ and $R_{BB} = R_1 \parallel R_2 = \frac{R_1 R_2}{R_1 + R_2} \rightarrow (2)$

$$V_{BB} - R_{BB} I_B - 0.7 - R_L I_E = -5$$

- And since $I_B = \frac{I_E}{\beta + 1} \rightarrow (3)$:

$$V_{BB} - R_{BB} \frac{I_E}{\beta + 1} - 0.7 - R_L I_E = -5$$

$$I_E = \frac{5 - 0.7 + V_{BB}}{R_L + \frac{R_{BB}}{\beta + 1}} \rightarrow (4)$$



Calculations

Step 3) Find Thevenin equivalent

For the BJT topology,

- if R_{BB} is large, it greatly reduces the value of I_E
- Due to equation 3, it also reduces I_B .
- If I_B becomes too small, the transistor may go into cutoff.
- Thus, R_{BB} cannot be large, so to make sure of this we want: $R_L \approx \frac{R_{BB}}{\beta+1}$, where $100 \leq \beta \leq 300$

$$\therefore R_{BB} \approx 10k\Omega$$



Calculations

Step 3) Find Thevenin equivalent

Now find V_{BB} :

- Recall step 1: $I_C = 24.75mA$
- Since $I_C \approx I_E$, plugging this into Eq. 4 gives:

$$I_C \approx I_E = \frac{5 - 0.7 + V_{BB}}{R_L + \frac{R_{BB}}{\beta + 1}}$$

$$24.75 = \frac{5 - 0.7 + V_{BB}}{0.1 + 0.1}$$

$$\therefore V_{BB} = 0.65V$$



Calculations

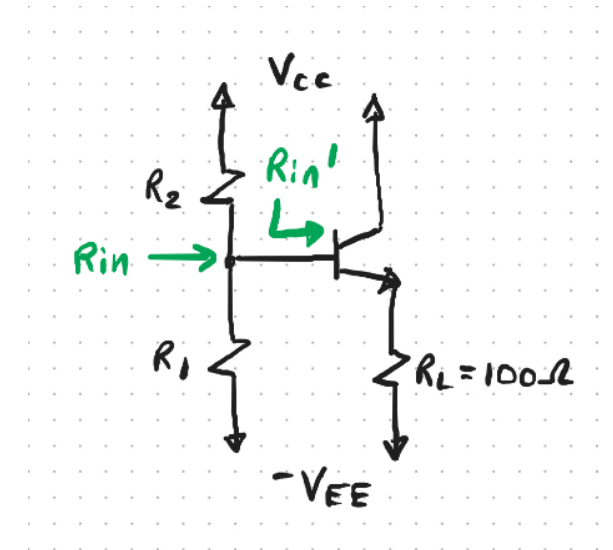
Step 4) Find R_{in}

$$R_{in} = R_1 \parallel R_2 \parallel R'_{in}, \text{ where } R'_{in} = (\beta + 1)(r_e + R_L)$$

$$R_{in} = R_{BB} \parallel (\beta + 1) \left(\frac{\alpha}{g_m} + R_L \right), \text{ where } \alpha = \frac{\beta}{\beta + 1}$$

$$R_{in} = 10k\Omega \parallel (100 + 1) \left(\frac{100/101}{0.9mS} + 100\Omega \right)$$

$$\therefore R_{in} = 5.0522k\Omega$$



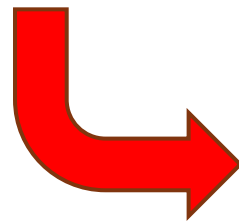
Calculations

Step 5) Determine Gain

$$G = \frac{R_{in}}{R_{in} + R_{sig}} \times \frac{R_L}{r_e + R_L}, \text{ where } r_e = \frac{\alpha}{g_m}$$

$$G = \frac{5.0522k\Omega}{5.0522k\Omega + 0.1k\Omega} \times \frac{100\Omega}{\frac{100/101}{990mS} + 100\Omega}$$

$$\therefore G = 0.9708 \checkmark\checkmark$$



$$0.9V \leq Gain \leq 1V$$



Calculations

Step 6) Validate Linearity

$$|V_{sig}| \frac{R_{in}}{R_{in} + R_{sig}} \times \frac{r_e}{r_e + R_L} \leq 0.2x$$
$$(0.5V) \frac{5.0522k\Omega}{5.0522k\Omega + 0.1k\Omega} \times \frac{\frac{100}{101}}{\frac{100}{101} + 100\Omega} \leq 0.2(25mV)$$

$$\therefore 0.005334 \leq 0.005 \checkmark\checkmark$$

- The result slightly exceeds the allowed limit. In practical design, this is acceptable.



Calculations

Step 7) Find resistor values that obtain the calculated bias current

- Plug R_{BB} and V_{BB} into Equations 1 and 2 respectively:

$$R_{BB} = \frac{R_1 R_2}{R_1 + R_2} = 10k\Omega$$

$$V_{BB} = 0.65 = -5 + 10 \left(\frac{R_2}{R_1 + R_2} \right)$$

$$\frac{R_2}{R_1 + R_2} = \frac{10k\Omega}{R_1} \rightarrow (5)$$

$$\frac{5.65}{10} = \frac{R_2}{R_1 + R_2} \rightarrow (6)$$



Calculations

Step 7) Find resistor values that obtain the calculated bias current

- Equating Equations 5 and 6:

$$0.565 = \frac{10k\Omega}{R_1}$$

$$\therefore R_1 = 17.699k\Omega$$

- Plugging R_1 into Equation 5 gives:

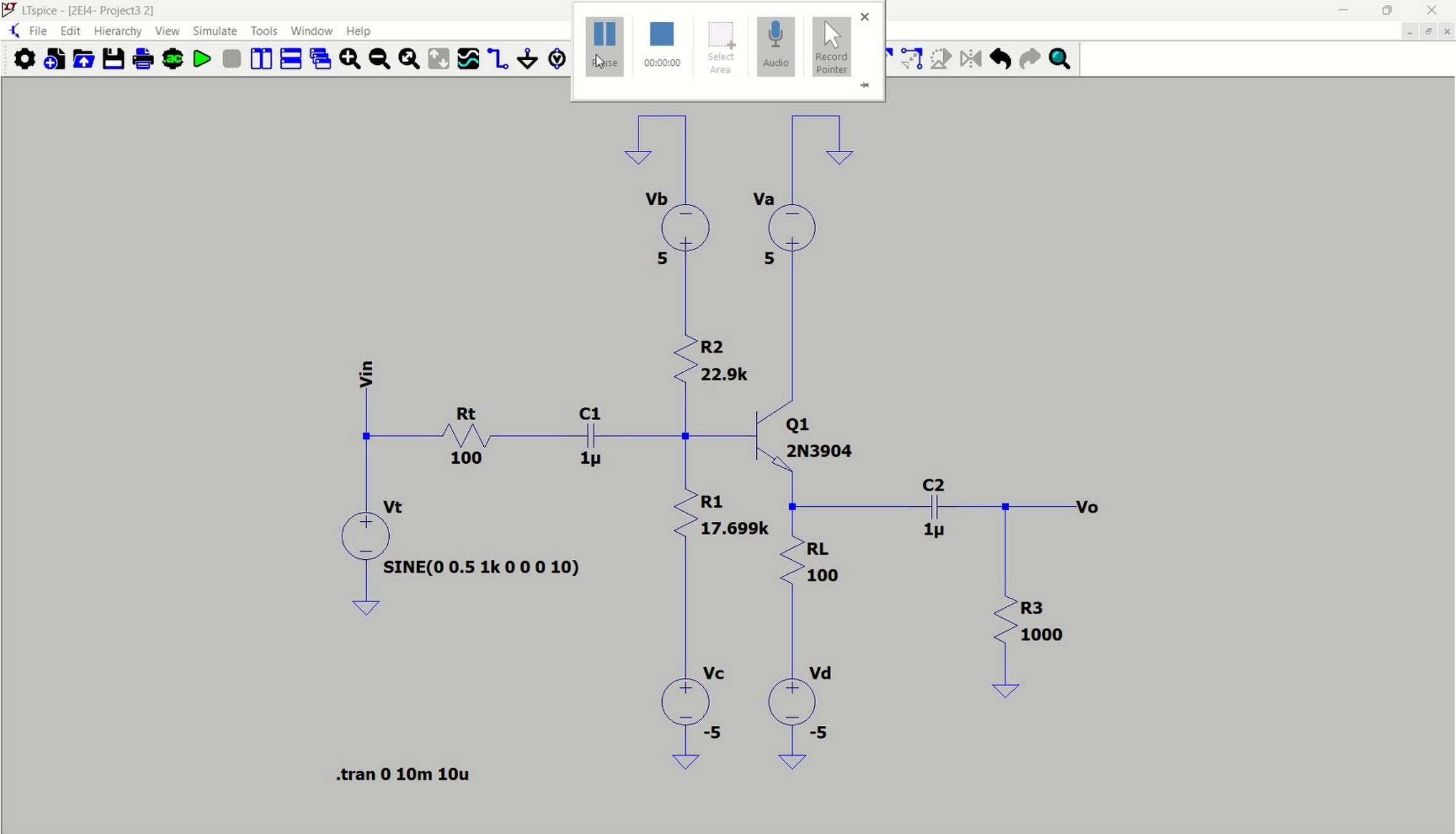
$$R_2 = 22.988k\Omega$$

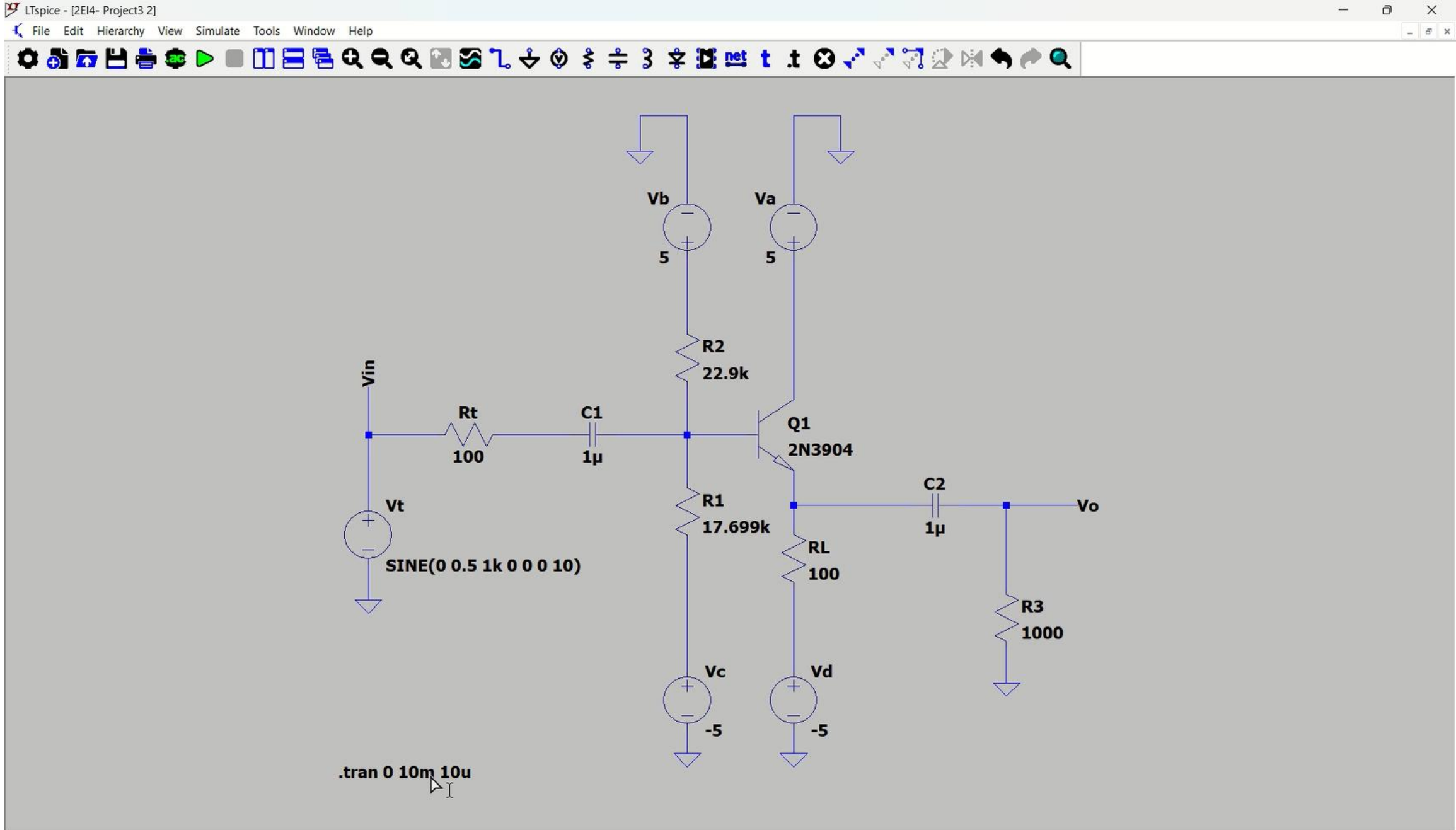


Part 2: Simulation

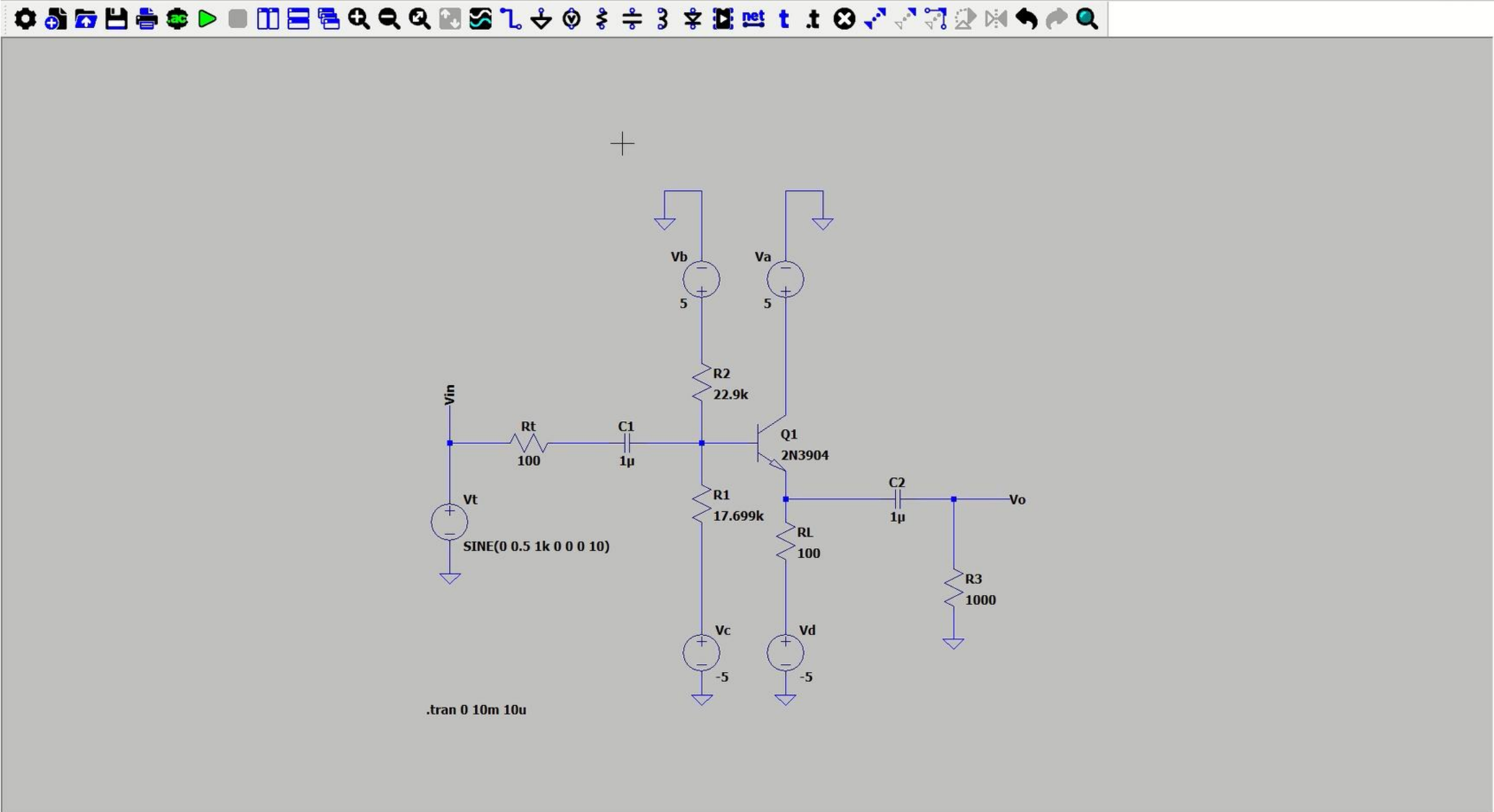
LT Spice







Right click to edit ".tran 0 10m 10u"



Other Performance Parameters

- An important parameter is the internal resistance R_{in} .
- A high input resistance is crucial for minimizing signal loss.
- Recall:

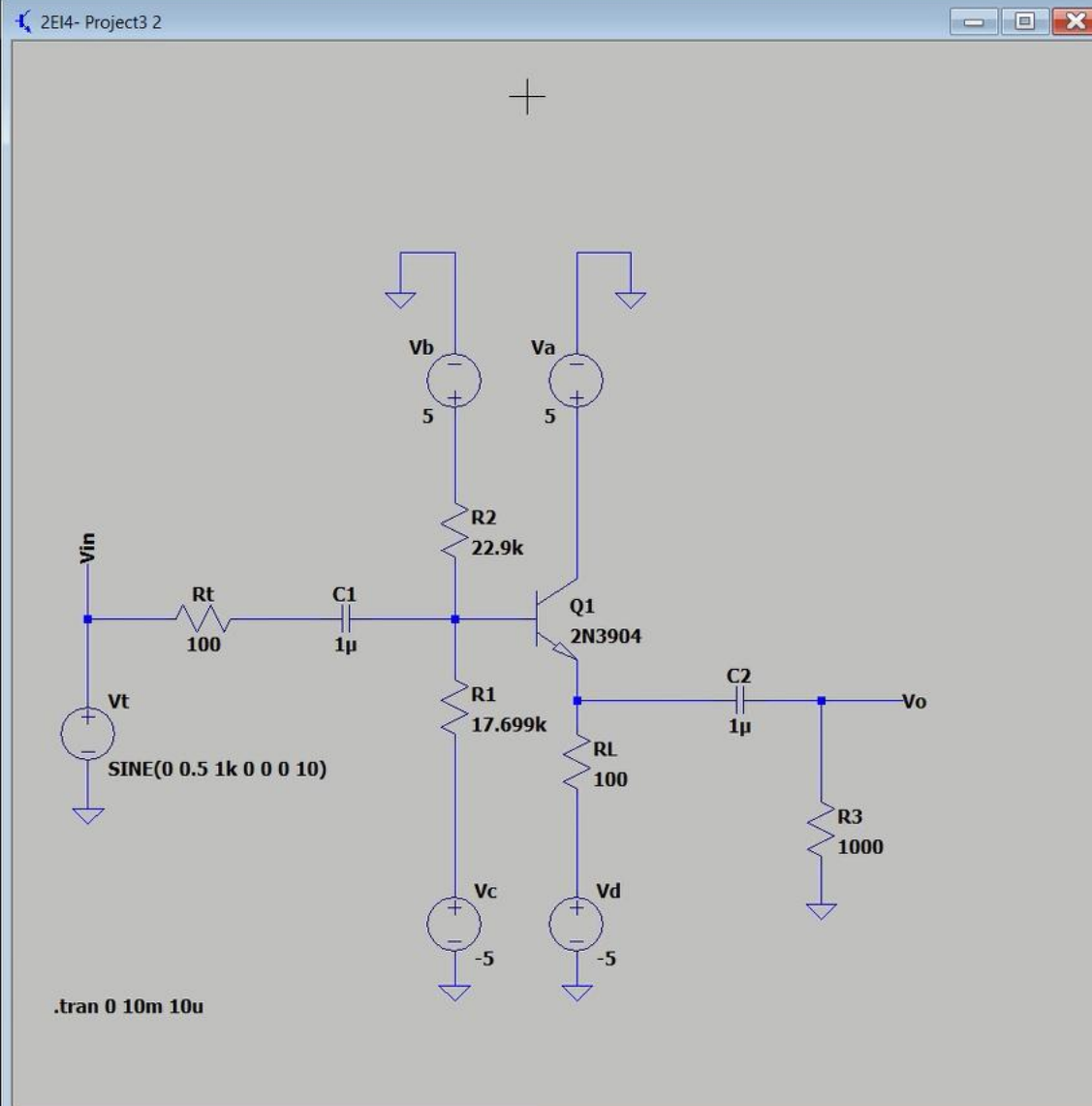
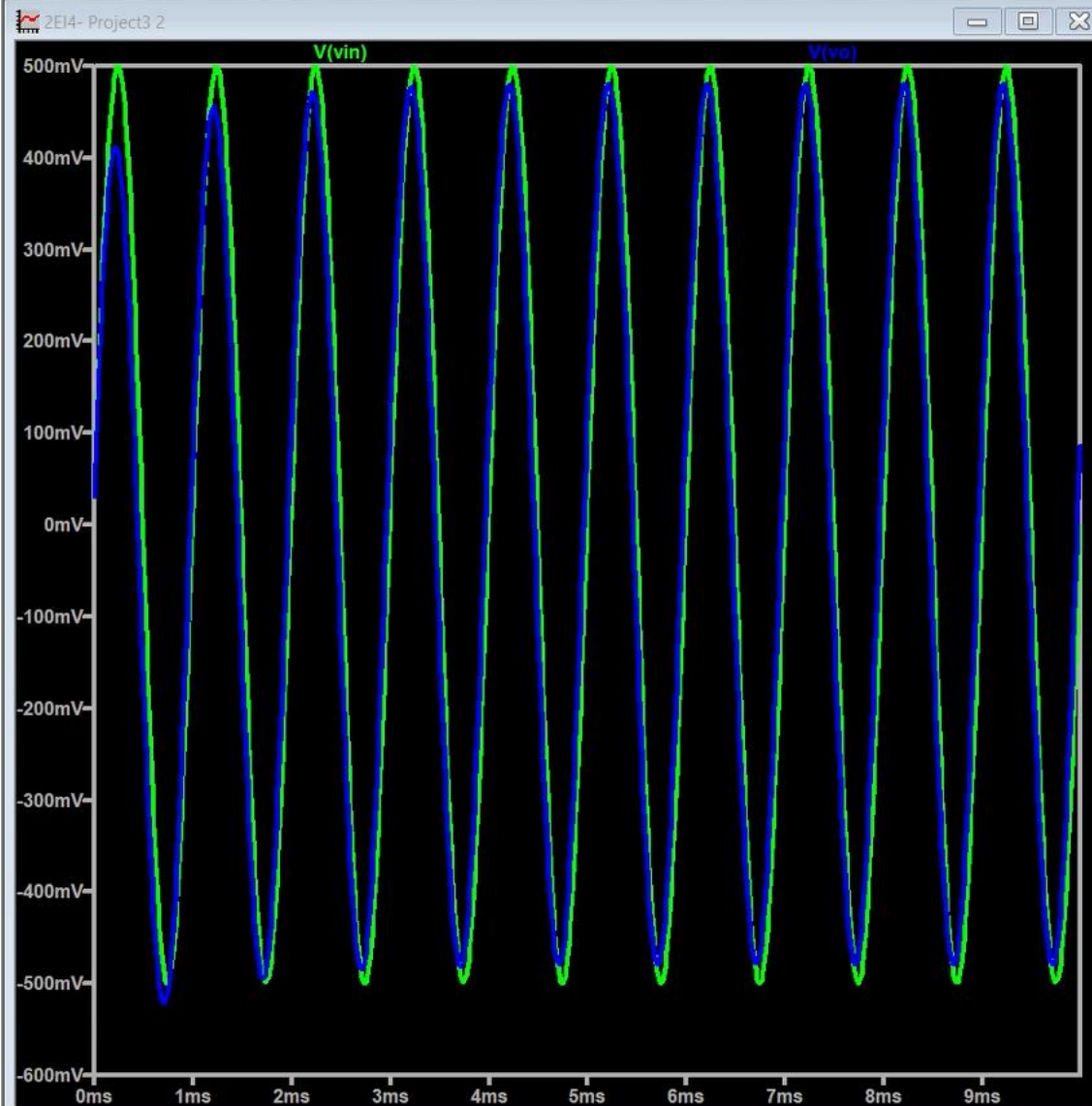
$$R_{in} = \mathbf{R_1} \parallel \mathbf{R_2} \parallel R'_{in} , \text{ where } R'_{in} = (\beta + 1)(r_e + R_L)$$

- Using large resistor values dominate the equation above => R_{in} is also large.
- If R_{in} is too low, the amplifier attenuates the input signal, lowering the gain.





2EI4- Project3 2 2EI4- Project3 2

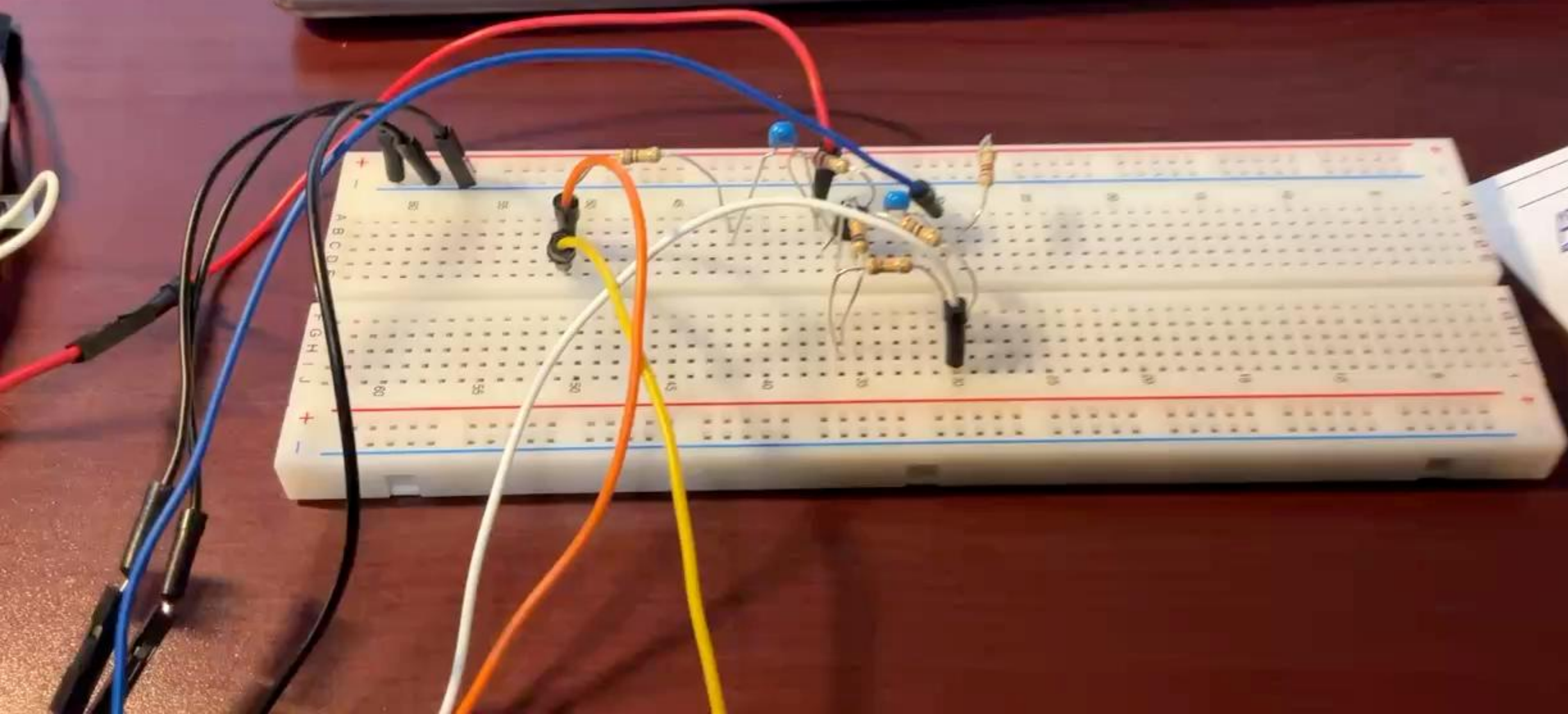


Part 3: Physical

Circuit and Measurement Performed



AMD
RADEON
GRAPHICS



Master Enable is Off

Hardware Power Limit: Auto 2.5 W

Positive Supply (V+) Rdy

5 V

- .----

☐ Tracking

Negative Supply (V-) Rdy

-5 V

- .----

System Monitor

Temperature: 47.69 °C / 117.84 °F

USB Voltage: 4.929 V

USB Current: 657.1 mA

V+: 0.070 V

Die: 61.95 °C / 143.51 °F

AUX Voltage: 0.037 V

AUX Current: 0.0 mA

V-: 0.022 V

